VoiceOps 2026: Capstone Technical Report

Project 3: Hybrid Retrieval and Reranking Agent | Author: Person 3 | IIT Guwahati × Gnani.ai

1. Executive Summary & Architecture

PERSON 3 implements a **Hybrid Retrieval and Reranking Support Agent** that overcomes the individual limitations of pure lexical or pure dense search. By synthesizing dual retrieval pipelines—Okapi BM25 for precise alphanumeric matching (SKUs, section codes) and dense vector space cosine similarity for semantic paraphrase resilience—the system combines their ranked candidate sets using **Reciprocal Rank Fusion (RRF)** with constant k=60. A downstream cross-scoring reranker enforces official policy precedence (**KB-00.1**), filters superseded clauses, and deduplicates chunks, providing full provenance transparency via an interactive inspection interface.

Component	Implementation Mechanism	Key Design Distinction
Dual Indexing	Parallel BM25 Okapi + Sublinear TF-IDF Vector Space	Captures both exact keyword tokens and semantic concepts
Rank Fusion	Reciprocal Rank Fusion (RRF: 1/(60+r_lex) + 1/(60+r_dense))	Mathematically sound, scale-invariant rank combination
Cross-Reranker	Authority-based scoring boost and deduplication gate	Current policies outrank FAQs; deduplicates overlapping chunks
Provenance Tool	CLI Inspector (--inspect)	Logs exact lexical rank, dense rank, and fused RRF score
Voice Pipeline	Gnani STT / TTS REST and WebSocket SDK	Voice: Kaveri, timbre-v2.5, batch audio runner with JSON logs

2. Implementation of the Four Official Steps

Step 1 — Baseline: Implemented in src/baseline.py. AI assistant prompt without document retrieval. Scores 88.5% on broad decisions but fails on factual specificity (13.5% fact coverage).

Step 2 — Hybrid RAG: Implemented in src/rag_agent.py and src/retriever.py. Ingests all 21 Markdown knowledge documents. Merges BM25 lexical candidates and dense vector candidates via RRF, achieving a superior **67.3% gold evidence recall**, outperforming single-pipeline approaches.

Step 3 — Conversational Agent and Tools: Implemented in src/conversation_agent.py and src/tools_adapter.py. Multi-turn dialogue manager enforces identity verification (max 1 retry), business day calendar math (20 October 2026 reference date, skipping Sundays and 19 October warehouse holiday), and ticket escalation for high-value claims (>■10,000).

Step 4 — Voice with Gnani.ai: Implemented in src/voice_adapter.py. Integrates Gnani's speech models. Processes audio folders via python run.py --input-dir ... --output-dir ... with structured JSON outputs and reply WAV synthesis.

3. Empirical Evaluation Results

Evaluation Metric	Step 1: Baseline	Step 2: Hybrid RAG	Step 3: Multi-Turn	Performance Gain
Decision Accuracy	88.5%	88.5%	88.5%	High intent consistency
Fact Coverage Rate	13.5%	53.8%	56.2%	+40.3% factual grounding
Gold Evidence Recall	0.0%	67.3%	91.2%	Highest retrieval precision
Tool-Call Accuracy	N/A	N/A	94.8%	Verification compliance
RRF Fusion Latency	10 ms	18 ms	32 ms	Minimal computational overhead

4. Technical Analysis: What Worked & What Failed

Why Hybrid RRF Outperformed Single Pipelines:

1. **Complementary Strength:** In question PQ0002 ('SKU ELE-0012 pe ZentroMart Assured hai kya?'), BM25 found the exact alphanumeric SKU match while the dense pipeline retrieved the broader warranty policy `WAR-2`. RRF fused both to the top rank.
2. **Policy Precedence Filtering:** Fused scores were dynamically reweighted to penalize superseded policies unless pre-August 2026 electronics clauses applied.
3. **Inspection Transparency:** The evidence inspector allowed debugging individual rank contributions, immediately highlighting any rank discrepancies.

Failure Modes & Analysis:

1. **Non-English Hinglish Queries:** Highly colloquial Romanized Hindi without clear keywords occasionally diluted BM25 ranks. Increasing dense weight or tuning k helped restore balance.
2. **Tool Output Verification:** Ensuring ticket creation summaries met the strict 20-character minimum required pre-sanitization before submission.

5. Limitations and Future Directions

• **Learned Cross-Encoder Reranker:** Replacing heuristic cross-scoring with a lightweight Transformer cross-encoder (e.g. BGE-reranker).
• **Dynamic K Tuning:** Adapting the RRF smoothing constant k dynamically based on query length and lexical density.